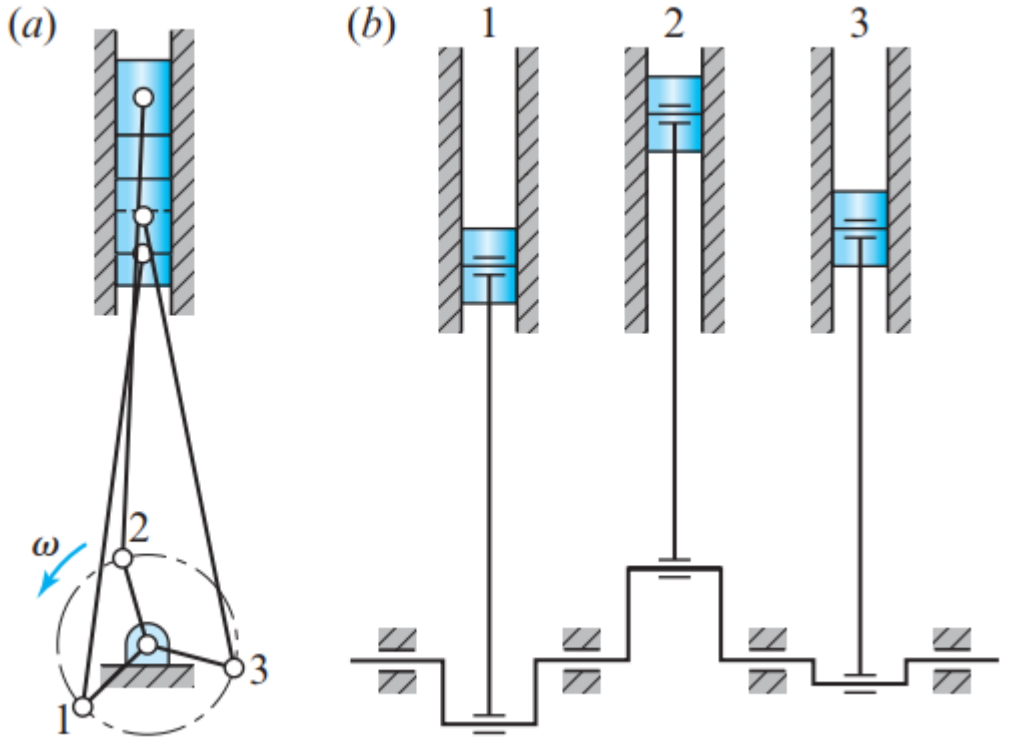
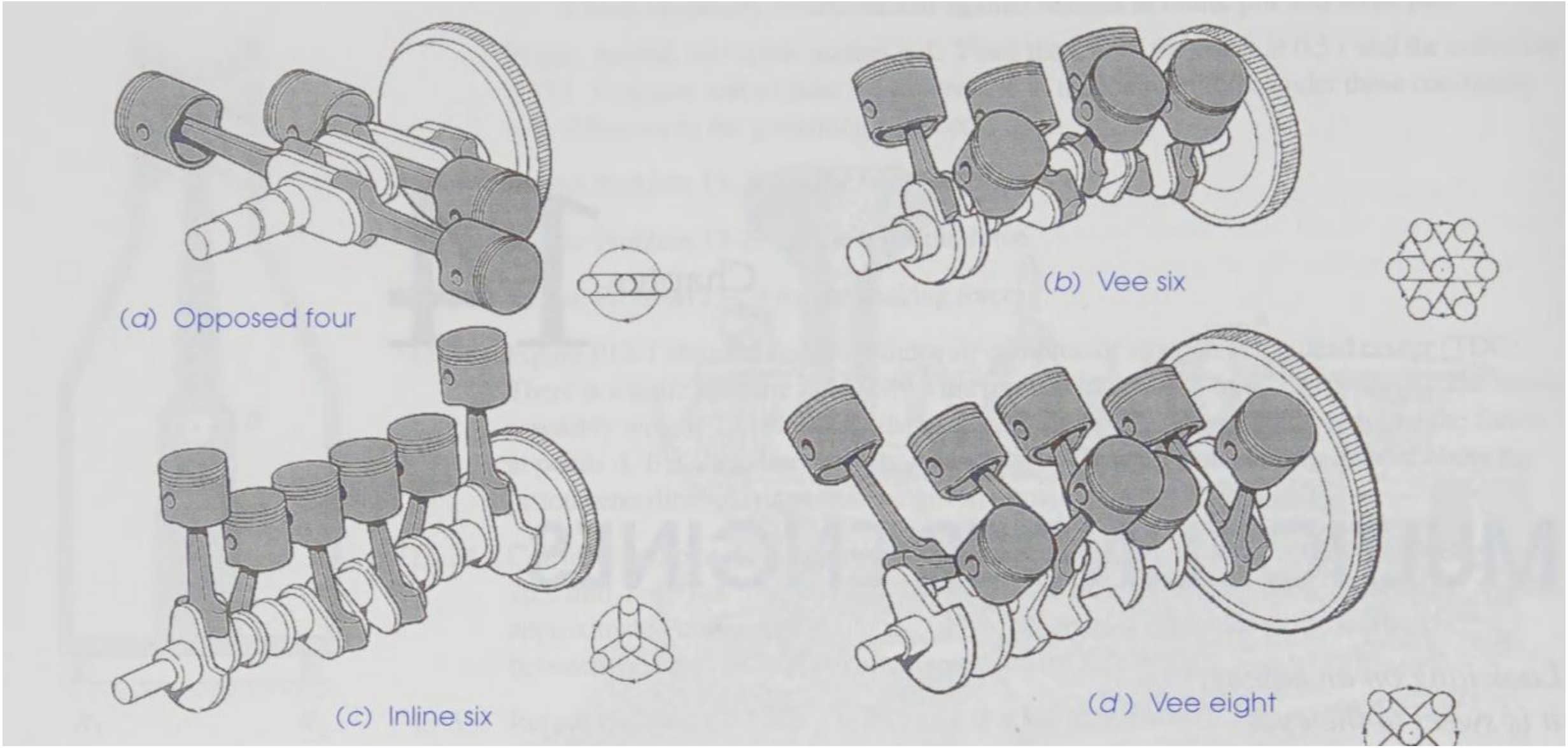
Balancing of multi-cylinder machines – engines/ compressors etc.

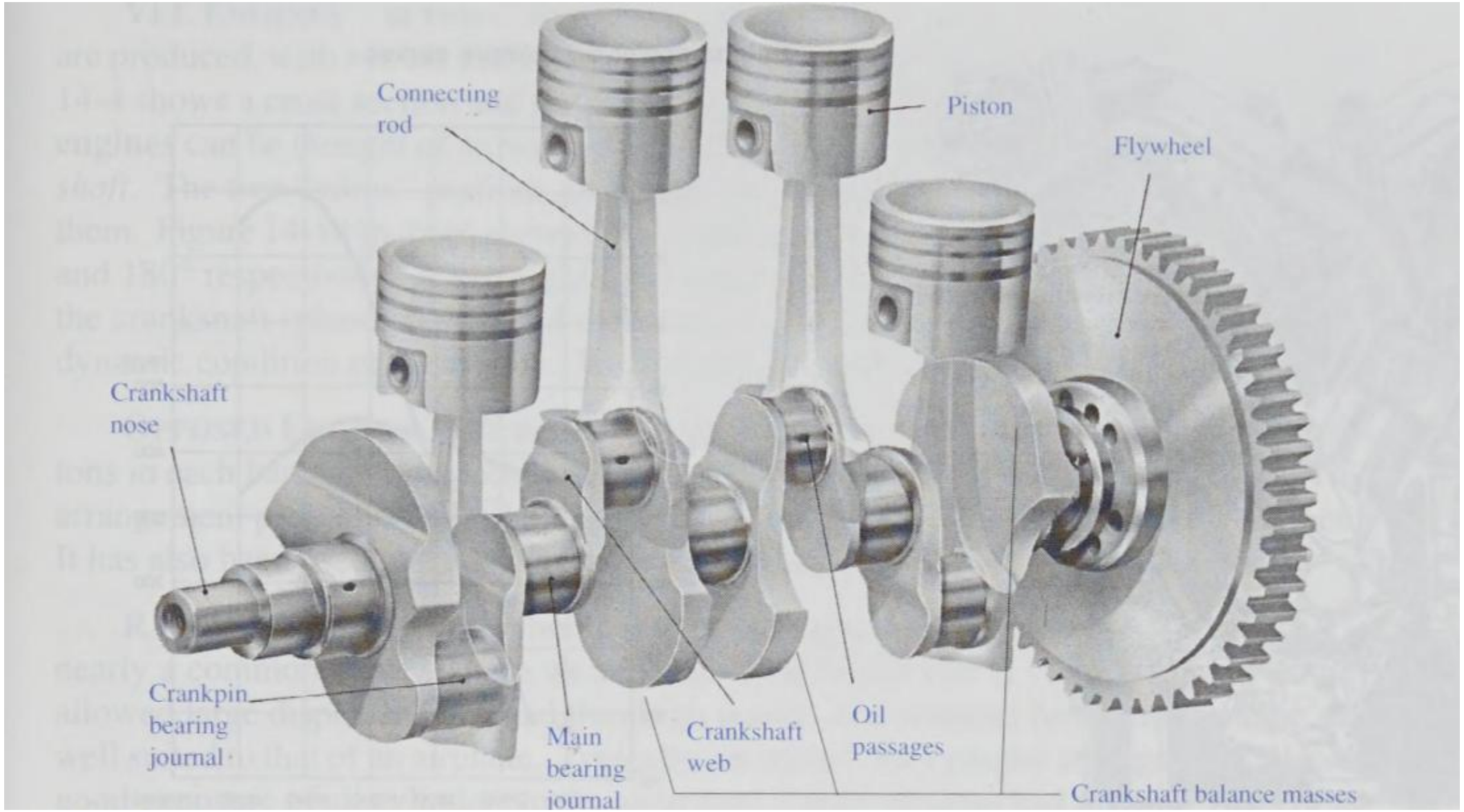


Three cylinder **in-line engine**

Balancing of multi-cylinder machines – engines/ compressors etc.

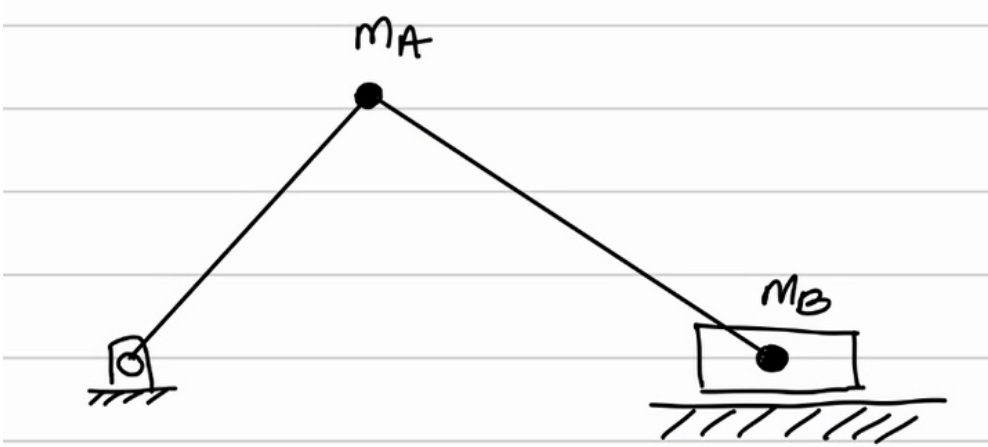


Balancing of multi-cylinder machines – engines/ compressors etc.



Balancing of multi-cylinder machines – engines/ compressors etc.

In multi-cylinder engines and compressors, the weight of the crankshaft and the part of the weight of the connecting rod that is concentrated at the crank pin can be counter-balanced exactly. Therefore, only the balancing of the reciprocating weight will be considered here.



$$\Rightarrow \vec{f_S} = \vec{f_A} + \vec{f_B} = R\omega^2 \left[\left(m_A + m_B - m_{cb} \right) \cos\theta + m_B \frac{R}{L} \cos 2\theta \right] \hat{i}$$

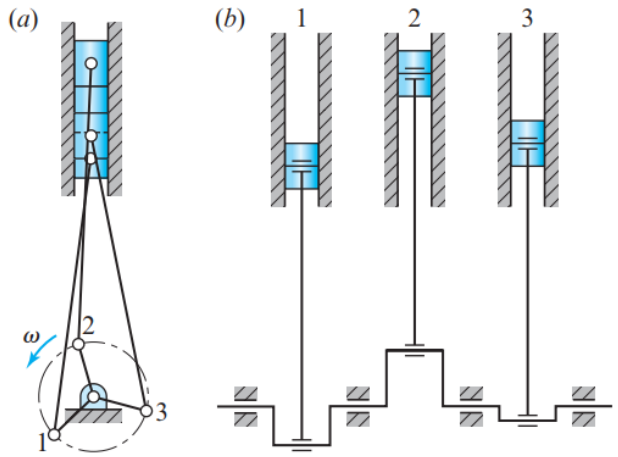
$$+ R\omega^2 \left(m_A - m_{cb} \right) \sin\theta \, \hat{j}$$

Balancing of multi-cylinder in-line engines

$$f_{B_i} = m_B R \omega^2 \left[\cos(\theta + \phi_i) + \frac{R}{L} \cos 2(\theta + \phi_i) \right], \quad i = 1, 2, \cdots, n$$

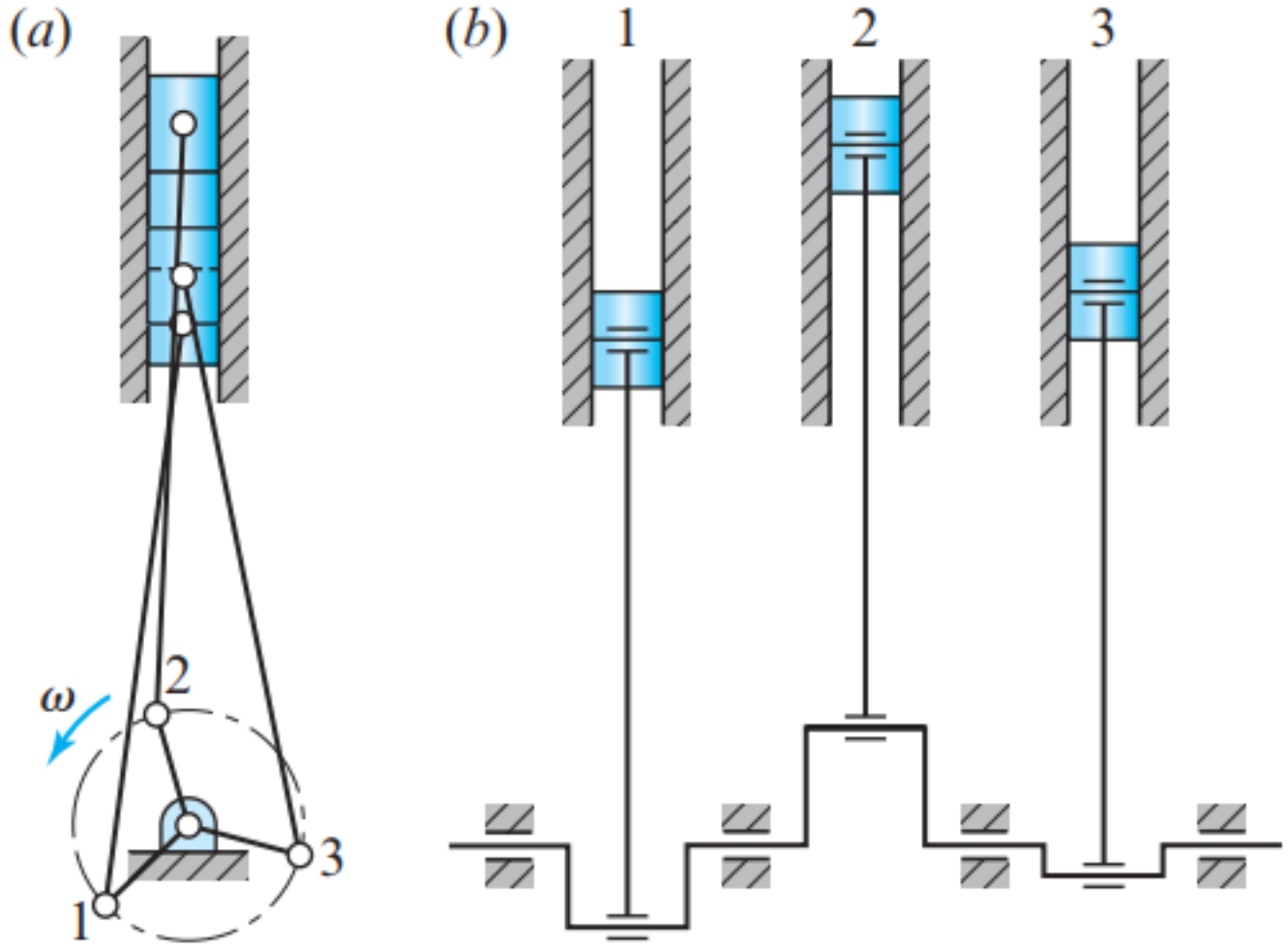
$\phi_i \rightarrow$ phase angle of the ith reciprocating mass.

$\theta \rightarrow$ crank angle (for the first cylinder).



Three cylinder **in-line engine**

Example– balancing of three-cylinder in-line engine



Example– balancing of three-cylinder in-line engine

Phase angles $\phi_2 = 120°$, $\phi_2 = 240°$ ($\phi_1 = 0$)

$$f_{B_1} + f_{B_2} + f_{B_3} = R\omega^2 \left(\cos\theta + \cos(\theta+120°) + \cos(\theta+240°)\right)$$

$$+ R\omega^2 \frac{R}{L} \left(\cos 2\theta + \cos 2(\theta+120°) + \cos 2(\theta+240°)\right)$$

$$\Rightarrow \cos\theta + \cos(\theta+120°) + \cos(\theta+240°) = \cos(\theta) + 2\cos\left(\frac{2\theta+360°}{2}\right)\cos\left(\frac{120°}{2}\right)$$

$$= \cos\theta + 2\cos(\theta+180°)\cos 60° = \cos\theta - \cos\theta = 0$$

Example– balancing of three-cylinder in-line engine

Similarly,

$$\cos 2\theta + \cos 2(\theta + 120^\circ) + \cos 2(\theta + 240^\circ) = 0$$

Using the identity $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

For 3-cylinder in-line engine, the first and second harmonics are inherently balanced.

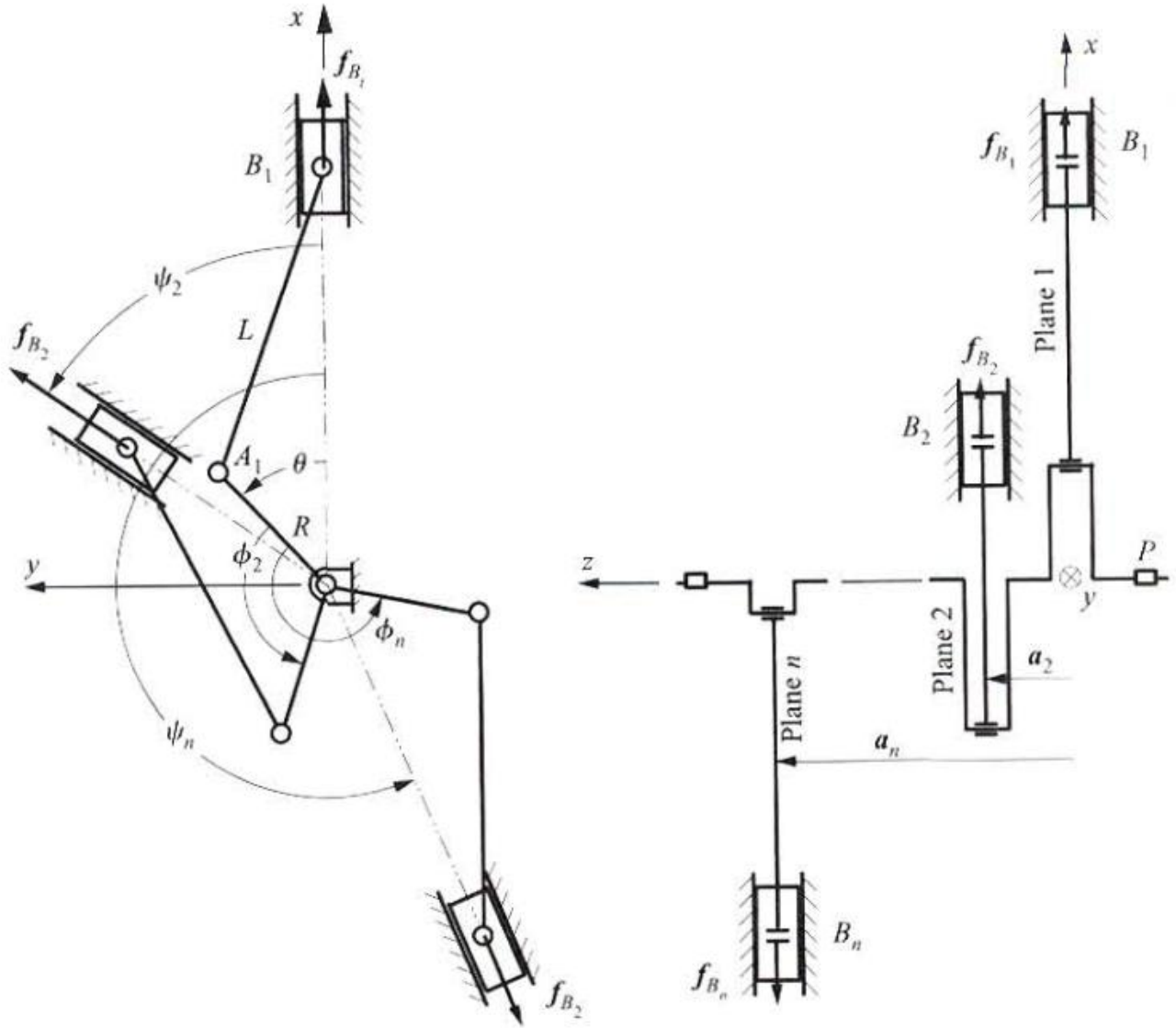


FIGURE 15.14 Schematic diagram of shaking forces in a multicylinder engine.

Inertia force for non-inline engines

$$f_B = m_B R\omega^2\left[\cos(\theta+\phi_i-\psi_i)+\frac{R}{L}\cos 2(\theta+\phi_i-\psi_i)\right],\quad i=1,2,\ldots,n$$

$\phi_i \rightarrow$ phase angle / angle of crank from the first crank $(\phi_1=0)$.

$\psi_i \rightarrow$ angle the cylinder axis makes with the x-axis.

The x-component of the shaking force from the ith cylinder is

$$\Rightarrow f_{Bx_i} = f_{B_i} \cos\psi_i = m_B R\omega^2 \left[\cos(\theta+\phi_i-\psi_i) + \frac{R}{L}\cos 2(\theta+\phi_i-\psi_i)\right]\cos\psi_i$$

the y component is

$$\Rightarrow f_{By_i} = f_{B_i} \sin\psi_i = m_B R\omega^2 \left[\cos(\theta+\phi_i-\psi_i) + \frac{R}{L}\cos 2(\theta+\phi_i-\psi_i)\right]\sin\psi_i$$

$$f_x = m_B R\omega^2 \left[\sum_{i=1}^{n} \cos(\theta + \phi_i - \psi_i) \cos\psi_i + \frac{R}{L} \sum \cos 2(\theta + \phi_i - \psi_i) \cos\psi_i \right]$$

and

$$f_y = m_B R\omega^2 \left[\sum_{i=1}^{n} \cos(\theta + \phi_i - \psi_i) \sin\psi_i + \frac{R}{L} \sum_{i=1}^{n} \cos 2(\theta + \phi_i - \psi_i) \sin\psi_i \right]$$

using, $\cos(\theta + \phi_i - \psi_i) = \cos\theta \cos(\phi_i - \psi_i) - \sin\theta \sin(\phi_i - \psi_i)$

To balance the forces, it is necessary for f_x and f_y to be zero.

This happens for all values of θ when

$$\sum_{i=1}^{n} \cos(\phi_i - \psi_i) \cos\psi_i = 0$$

$$\sum_{i=1}^{n} \cos(\phi_i - \psi_i) \sin\psi_i = 0$$

$$\sum_{i=1}^{n} \sin(\phi_i - \psi_i) \sin\psi_i = 0$$

$$\sum_{i=1}^{n} \sin(\phi_i - \psi_i) \cos\psi_i = 0$$

} For balancing primary shaking forces.

$$\sum_{i=1}^{n} \cos 2(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^{n} \cos 2(\phi_i - \psi_i) \sin \psi_i = 0$$

} For balancing secondary shaking forces.

$$\sum_{i=1}^{n} \sin 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^{n} \sin 2(\phi_i - \psi_i) \cos \psi_i = 0$$

For balancing momente

$$M_x = \sum_{i=1}^{n} f_{By_i} a_i = 0$$

$$M_y = \sum_{i=1}^{n} f_{Bx_i} a_i = 0$$

From this, we get

$$\sum_{i=1}^{n} a_i \cos(\phi_i - \psi_i) \cos\psi_i = 0$$

$$\sum_{i=1}^{n} a_i \cos(\phi_i - \psi_i) \sin\psi_i = 0$$

$$\sum_{i=1}^{n} a_i \sin(\phi_i - \psi_i) \sin\psi_i = 0$$

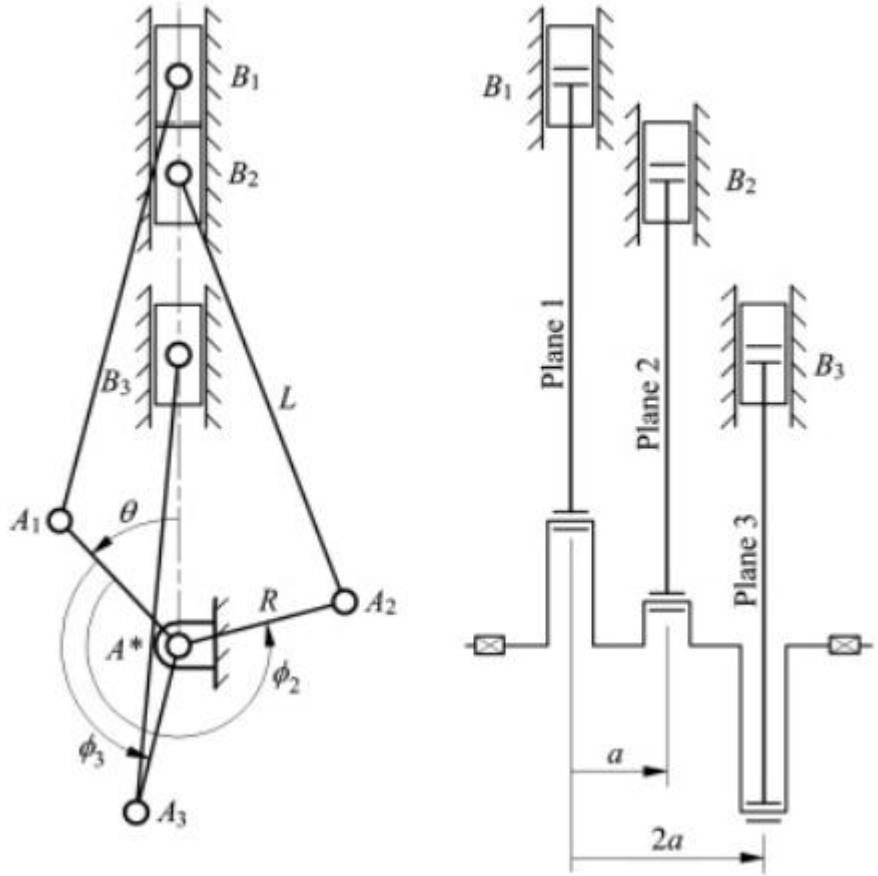
$$\sum_{i=1}^{n} a_i \sin(\phi_i - \psi_i) \cos\psi_i = 0$$

$$\sum_{i=1}^{n} a_i \cos 2(\phi_i - \psi_i) \cos \psi_i = 0$$

$$\sum_{i=1}^{n} a_i \cos 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^{n} a_i \sin 2(\phi_i - \psi_i) \sin \psi_i = 0$$

$$\sum_{i=1}^{n} a_i \sin 2(\phi_i - \psi_i) \cos \psi_i = 0$$


B_1
B_2
B_3
L
A_1
θ
R
A_2
A^*
ϕ_2
ϕ_3
A_3
B_1
B_2
B_3
Plane 1
Plane 2
Plane 3
a
$2a$

Moment components for the three-cylinder engine.

$$\psi_i = 0 \quad , \quad \sin \psi_i = 0$$

$$\sum_{i=1}^{3} a_i \cos \phi_i = 0 + a \cos 240^\circ + 2a \cos 120^\circ = 0 - \frac{a}{2} - \frac{2a}{2} = -\frac{3a}{2}$$

$$\sum_{i=1}^{3} a_i \sin \phi_i = 0 + a \sin 240^\circ + 2a \sin 120^\circ = 0 - \frac{a\sqrt{3}}{2} + \frac{2a\sqrt{3}}{2} = \frac{a\sqrt{3}}{2}$$

$$\sum_{i=1}^{3} a_i \cos 2\phi_i = 0 + a \cos 480^\circ + 2a \cos 240^\circ = 0 - \frac{a}{2} - \frac{2a}{2} = -\frac{3a}{2}$$

$$\sum_{i=1}^{3} a_i \sin 2\phi_i = 0 + a \sin 480^\circ + 2a \sin 240^\circ = 0 + \frac{a\sqrt{3}}{2} - \frac{2a\sqrt{3}}{2} = -\frac{a\sqrt{3}}{2}$$

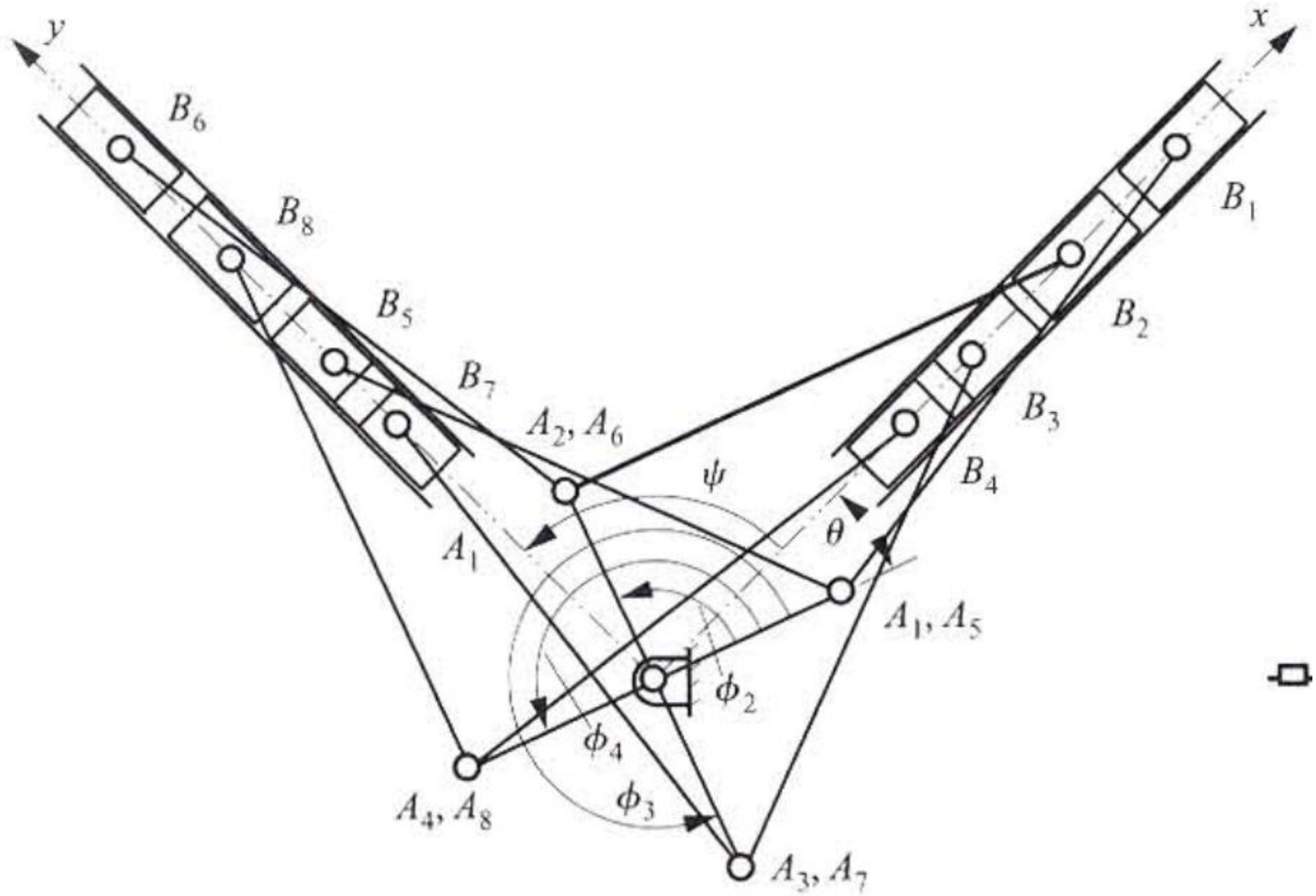
$$M = m_B R\omega^2 \left[-\frac{3a}{2}\cos\theta - a\frac{\sqrt{3}}{2}\sin\theta - \frac{R}{L}\frac{3a}{2}\cos 2\theta + \frac{R}{L}a\frac{\sqrt{3}}{2}\sin 2\theta \right]$$

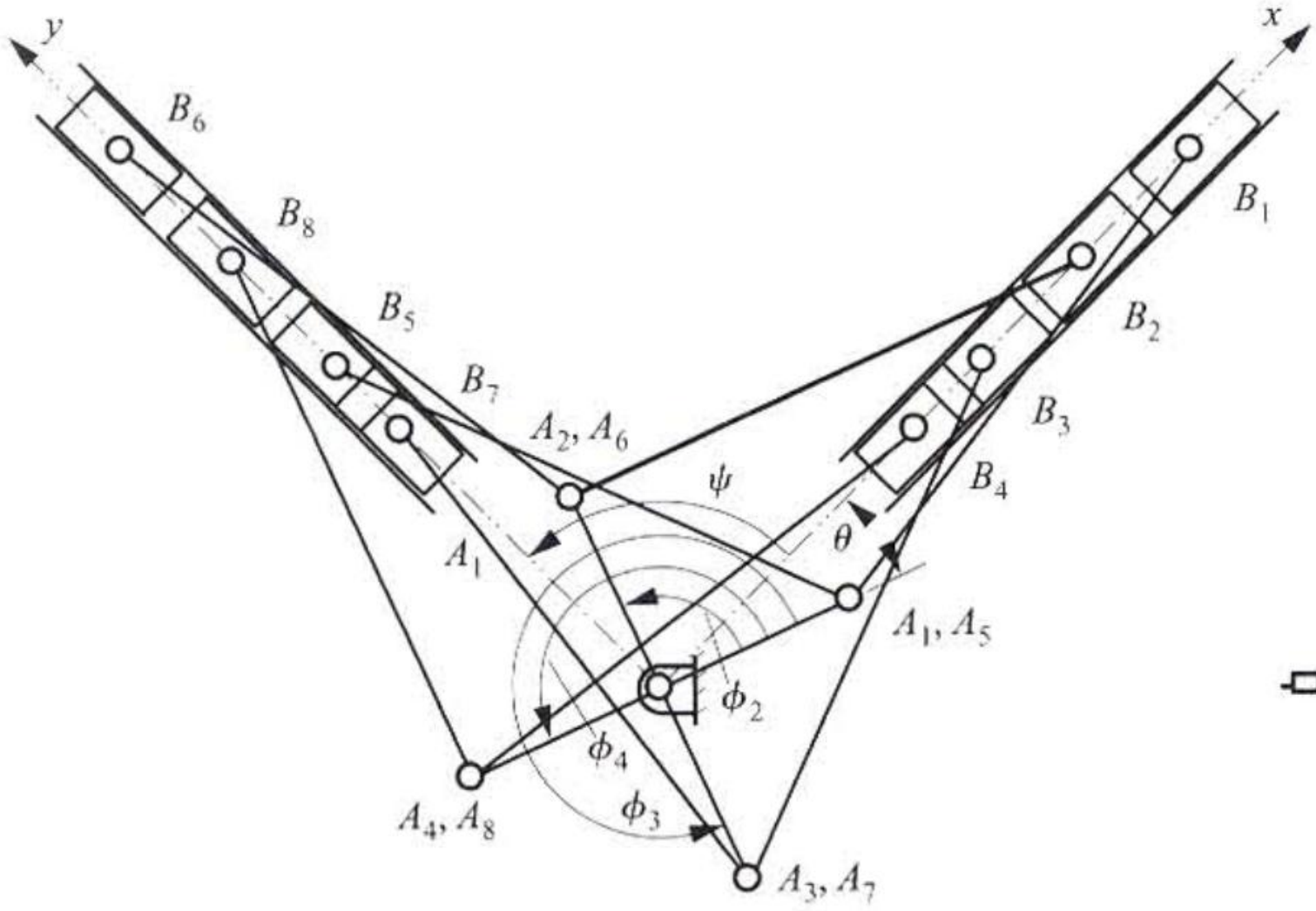
$$= m_B \frac{aR\omega^2}{2}\left[-3\cos\theta - \sqrt{3}\sin\theta - 3\frac{R}{L}\cos 2\theta + \frac{R}{L}\sqrt{3}\sin 2\theta \right]$$

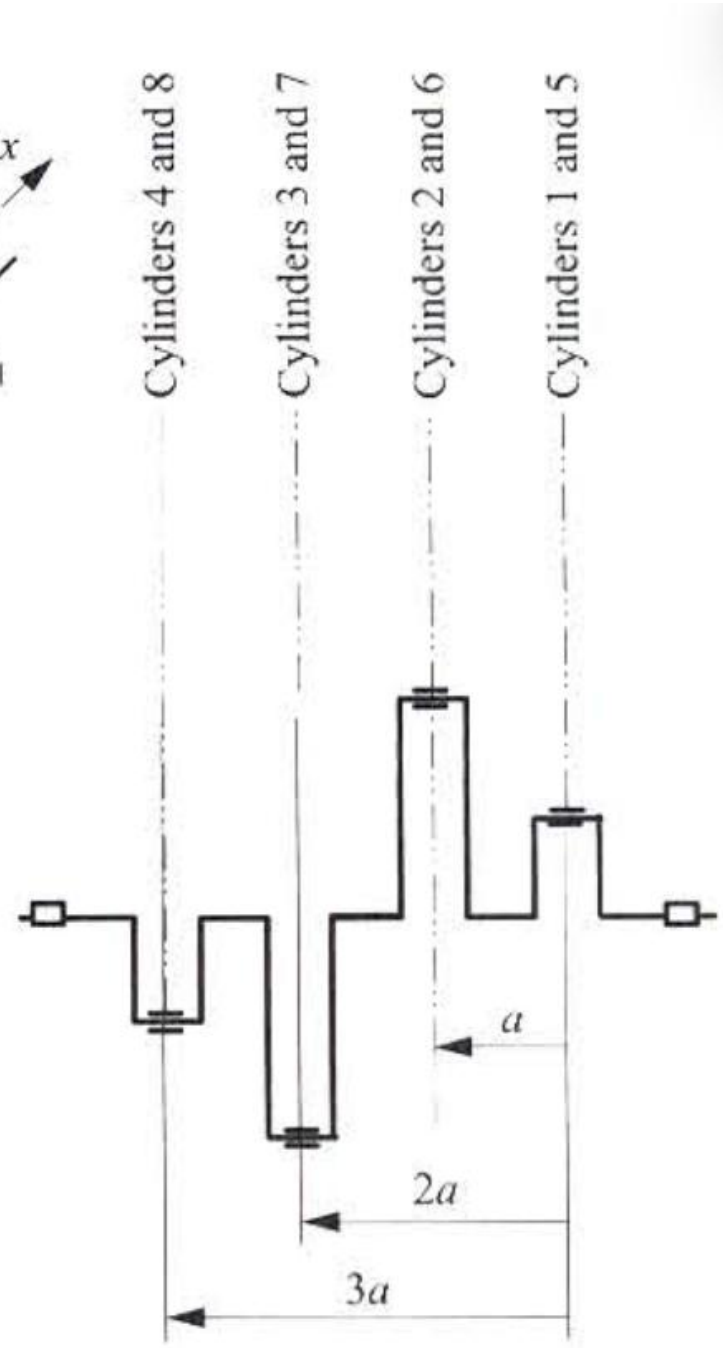
$$M = \frac{m_B a R\omega^2}{2}\left[-2\sqrt{3}\sin(\theta + 60^\circ) - 2\frac{R}{L}\sqrt{3}\sin(2\theta - 60^\circ) \right]$$

Primary shaking moment

secondary shaking moment


y
x
B_6
B_8
B_5
B_7
A_2, A_6
B_1
B_2
B_3
B_4
ψ
θ
A_1
A_1, A_5
ϕ_2
ϕ_4
ϕ_3
A_4, A_8
A_3, A_7


y
x
B_6
B_8
B_5
B_7
B_1
B_2
B_3
B_4
A_2, A_6
A_1
A_1, A_5
A_4, A_8
A_3, A_7
ψ
θ
ϕ_2
ϕ_4
ϕ_3


Cylinders 4 and 8
Cylinders 3 and 7
Cylinders 2 and 6
Cylinders 1 and 5
a
2a
3a